

**SIMATS ENGINEERING
ASSESSMENT TOOLS**

COURSE NAME: **Theory of Computation**
04

COURSE CODE: **CSA13**

COURSE OUTCOME COVERED

CO3: Construct regular expressions, and use the pumping lemma and closure properties to analyze regular languages. (BL:3)

Assessment Tool Weightage: 40%

QUESTIONS: 5
Duration : 1 Hour

Total Marks:100

Experiments

Code of Conduct:

I U.Divya/192372277 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Divya

SIMATS ENGINEERING ASSESSMENT TOOLS

Sl. No. Lab Experiments - Questions

- 1 Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)
 $S \rightarrow 0A1 \quad A \rightarrow 0A \mid 1A \mid \epsilon$
- 2 Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)
 $S \rightarrow 0S0 \mid 1S1 \mid 0 \mid 1 \mid \epsilon$
- 3 Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)
 $S \rightarrow 0S0 \mid A \quad A \rightarrow 1A \mid \epsilon$
- 4 Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)
 $S \rightarrow 0S1 \mid \epsilon$
- 5 Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)
 $S \rightarrow A101A \quad A \rightarrow 0A \mid 1A \mid \epsilon$

ANSWERS

1. Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)

$S \rightarrow 0A1 \quad A \rightarrow 0A \mid 1A \mid \epsilon$

AIM

To write a C program to check whether a given string belongs to the language defined by the CFG.

CODE

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main() {
```

```
    char str[100];
```

```
    int i, len, valid = 1;
```

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```
printf("Enter a binary string: ");

scanf("%s", str);

len = strlen(str);

if (len < 2 || str[0] != '0' || str[len - 1] != '1')

    valid = 0;

for (i = 0; i < len; i++) {

    if (str[i] != '0' && str[i] != '1') {

        valid = 0;

        break;

    }

}

if (valid)

    printf("String belongs to the CFG.\n");

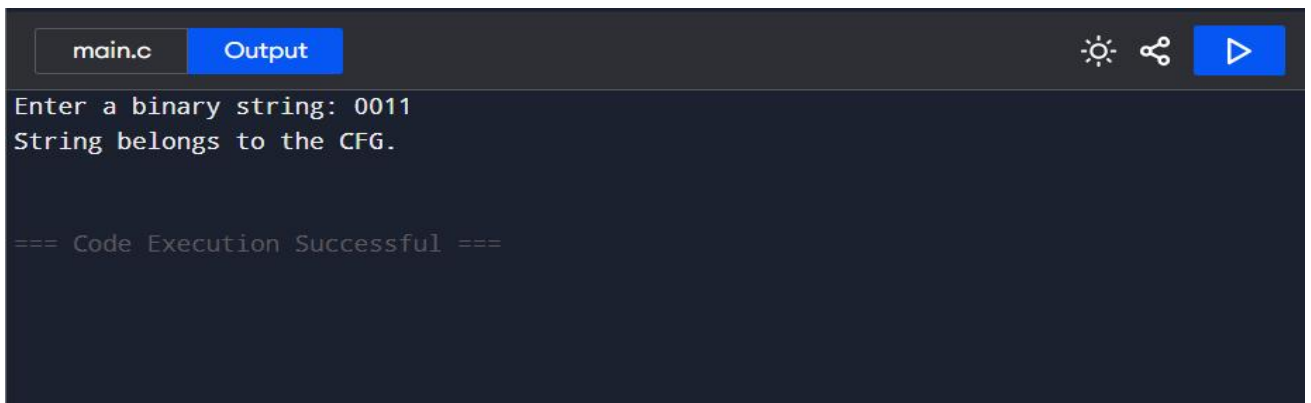
else

    printf("String does not belong to the CFG.\n");

return 0;

}
```

OUTPUT



```
main.c  Output

Enter a binary string: 0011
String belongs to the CFG.

=== Code Execution Successful ===
```

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2. Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)

$S \rightarrow 0S0 \mid 1S1 \mid 0 \mid 1 \mid \epsilon$

AIM

To write a C program to check whether a given string belongs to the language defined by the CFG.

CODE

```
#include <stdio.h>

#include <string.h>

int main() {

    char str[100];

    int i, j, len, valid = 1;

    printf("Enter a binary string: ");

    scanf("%s", str);

    len = strlen(str);

    for (i = 0; i < len; i++) {

        if (str[i] != '0' && str[i] != '1') {

            valid = 0;

            break;

        }

    }

    for (i = 0, j = len - 1; i < j && valid; i++, j--) {

        if (str[i] != str[j])

            valid = 0;

    }

}
```

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```
if (valid)

    printf("String belongs to the CFG.\n");

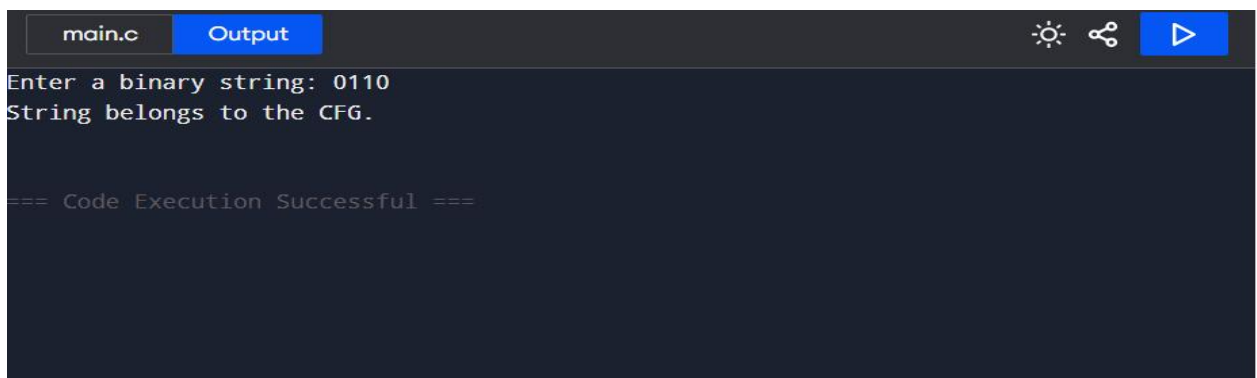
else

    printf("String does not belong to the CFG.\n");

return 0;

}
```

OUTPUT



The screenshot shows a code editor window with a tab labeled 'main.c' and an 'Output' pane. The output pane displays the following text: 'Enter a binary string: 0110', 'String belongs to the CFG.', and '=== Code Execution Successful ==='. The code editor has a dark theme and icons for settings, sharing, and running.

3. Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)

$S \rightarrow 0S0 \mid A$ $A \rightarrow 1A \mid \epsilon$

AIM

To write a C program to check whether a given string belongs to the language defined by the CFG.

CODE

```
#include <stdio.h>

#include <string.h>

int main() {

    char str[100];

    int i, len, left = 0, middle = 0, right = 0;

    int valid = 1;
```

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```
printf("Enter a binary string: ");

scanf("%s", str);

len = strlen(str);

i = 0;

while (i < len && str[i] == '0') {

    left++;

    i++;

}

while (i < len && str[i] == '1') {

    middle++;

    i++;

}

while (i < len && str[i] == '0') {

    right++;

    i++;

}

if (i != len || left != right)

    valid = 0;

if (valid)

    printf("String belongs to the CFG.\n");

else

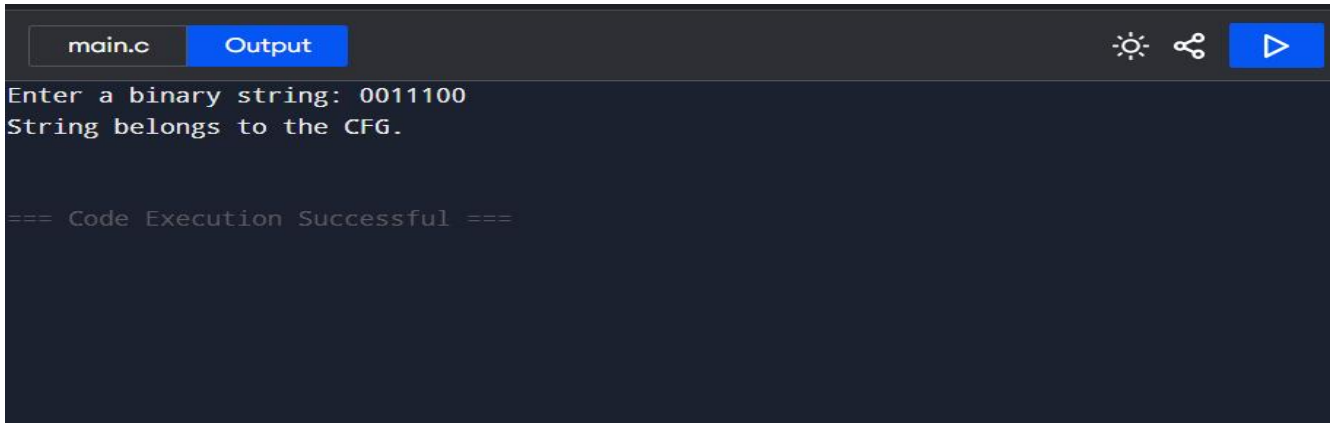
    printf("String does not belong to the CFG.\n");

return 0;
```

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}

OUTPUT



```
main.c  Output
Enter a binary string: 0011100
String belongs to the CFG.

=== Code Execution Successful ===
```

4. Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)

$S \rightarrow 0S1 \mid \epsilon$

AIM

To write a C program to check whether a given string belongs to the language defined by the CFG.

CODE

```
#include <stdio.h>

#include <string.h>

int main() {

    char str[100];

    int i, len, zeros = 0, ones = 0;

    int valid = 1;

    printf("Enter a binary string: ");

    scanf("%s", str);

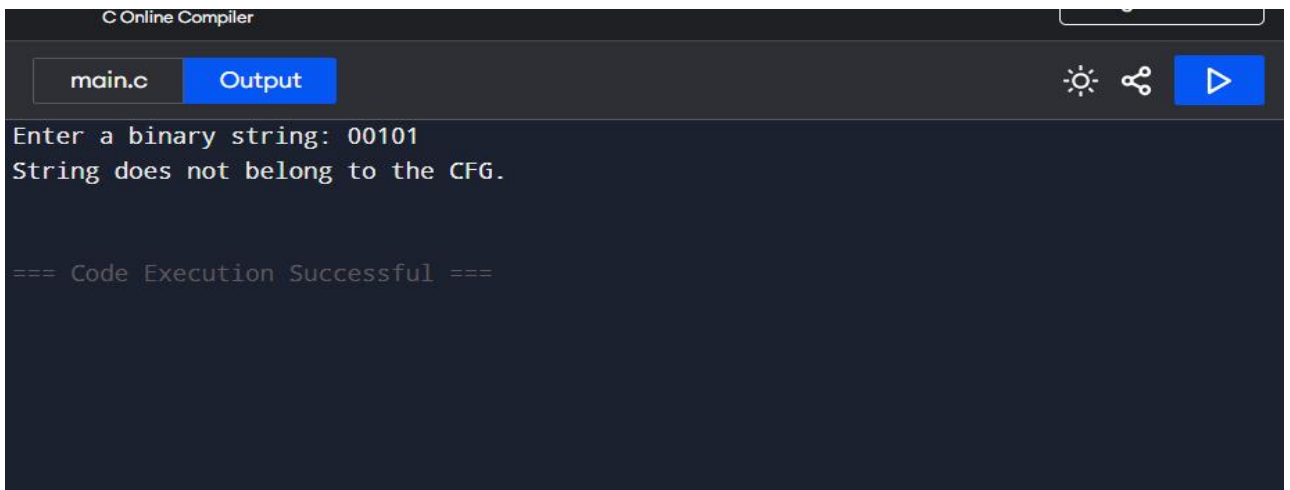
    len = strlen(str);

    i = 0;
```

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```
while (i < len && str[i] == '0') {  
    zeros++;  
    i++;  
}  
while (i < len && str[i] == '1') {  
    ones++;  
    i++;  
}  
if (i != len || zeros != ones)  
    valid = 0;  
if (valid)  
    printf("String belongs to the CFG.\n");  
else  
    printf("String does not belong to the CFG.\n");  
return 0;  
}
```

OUTPUT



The screenshot shows a web-based C compiler interface. At the top, it says "C Online Compiler". Below that, there are two tabs: "main.c" and "Output". The "Output" tab is active, showing the program's execution. The input prompt is "Enter a binary string: 00101". The output is "String does not belong to the CFG.". At the bottom, it says "=== Code Execution Successful ===".

```
C Online Compiler  
main.c Output  
Enter a binary string: 00101  
String does not belong to the CFG.  
=== Code Execution Successful ===
```


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5. Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)

$S \rightarrow A101A$ $A \rightarrow 0A \mid 1A \mid \varepsilon$

AIM

To write a C program to check whether a given string belongs to the language defined by the CFG.

CODE

```
#include <stdio.h>

#include <string.h>

int main() {

    char str[100];

    int i, len, found = 0, valid = 1;

    printf("Enter a binary string: ");

    scanf("%s", str);

    len = strlen(str);

    for (i = 0; i < len; i++) {

        if (str[i] != '0' && str[i] != '1') {

            valid = 0;

            break;

        }

    }

    if (valid) {

        for (i = 0; i <= len - 3; i++) {

            if (str[i] == '1' &&

                str[i + 1] == '0' &&

                str[i + 2] == '1') {
```

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```
        found = 1;

        break;
    }

}

}

if (valid && found)

    printf("String belongs to the CFG.\n");

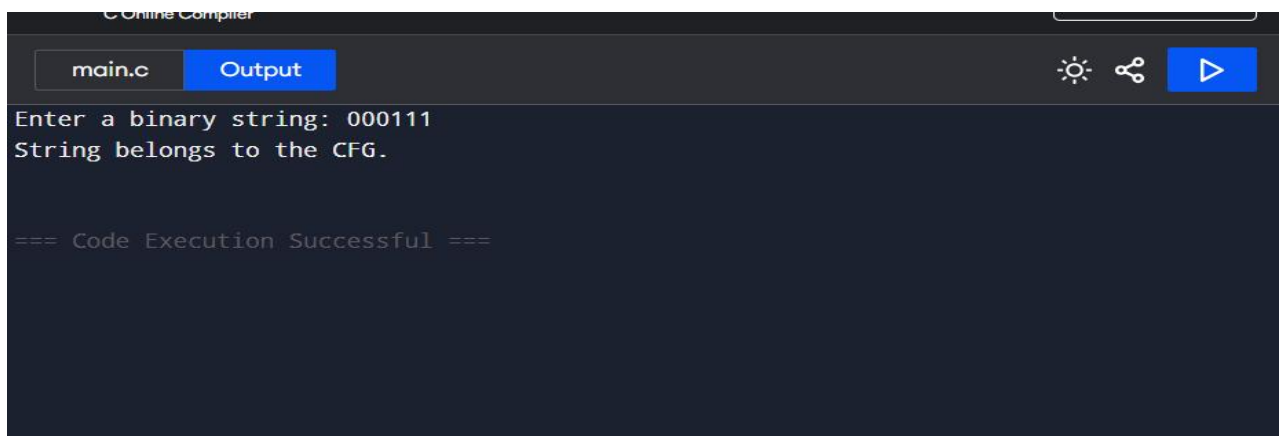
else

    printf("String does not belong to the CFG.\n");

return 0;

}
```

OUTPUT



The screenshot shows a web-based C compiler interface. At the top, there's a header "C Online Compiler". Below it, a dark-themed toolbar contains a file tab labeled "main.c", a blue "Output" button, and icons for settings, sharing, and a play button. The main area is a dark terminal window. It displays the prompt "Enter a binary string: 000111" followed by the output "String belongs to the CFG.". At the bottom of the terminal, it says "=== Code Execution Successful ===".

```
C Online Compiler
main.c Output
Enter a binary string: 000111
String belongs to the CFG.

=== Code Execution Successful ===
```

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RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand